WORKSHOP #2 (PART 1) - DESIGN ARTIFACTS AND SYSTEM MODELING

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1. System Domain

1.1 System Context

In Bogotá, citizens commonly perceive that public construction works miss their deadlines, and that information about their progress and effects on mobility is scattered or hard to find. This lack of accessible information reduces public trust in the district administration and limits citizens' ability to exercise oversight.

The Construction Works Oversight Website is a web platform that centralizes information about active public construction sites in Bogotá during 2026, with an initial scope of 200 construction sites. It presents each construction site on an interactive map together with its physical progress, performance indicators, budget and contractor and gives citizens tools to participate: rating and commenting on construction sites, comparing them, subscribing to updates, and preparing a right of petition.

System boundaries. The system *does*: display and consult construction site information; calculate performance indicators; manage user accounts and citizen participation; let the administrator manage construction site data and moderate content. The system *does not*: file rights of petition before public entities (it only provides a guide and a downloadable template); synchronize in real time with district information systems (data is loaded by the administrator from open data sources); or verify the legal identity of citizens.

External system	Interaction with the system
Open data sources (IDU, Secretaría de Movilidad de Bogota, SECOP II)	Provide the public information about construction sites (contract, budget, dates, progress, road closures), which the administrator loads into the system with its source and update date (NFR_14).

Email service	Sends password recovery messages (FR_18) and periodic updates to users subscribed to a construction site (FR_06).
Map provider	Supplies the base map of Bogotá on which construction sites and road closures are drawn (FR_01, FR_29).

1.2 Actor Characterization

Attribute	Visitor	Registered citizen	Administrator
Description	Any citizen who accesses the website without an account or without logging in.	Citizen who created an account with an email address and is logged in.	Member of the platform team responsible for the data and the content published on the site.
Goals	Find out which construction sites are active near them, their progress and road closures.	Participate in the oversight of construction sites: express opinions, follow specific sites and take formal action.	Keep construction site information accurate and up to date, and keep the discussion respectful and reliable.
Permissions	Read-only: map, detail sheet, search, filters, sorting, evolution graph, milestone timeline and road closures.	Everything a visitor can do, plus: rate, comment, delete own comments, report comments, compare, export to PDF, subscribe, download the	Add, edit and delete construction sites; delete any comment; access the administrator dashboard. Account is assigned by the system, not created

		petition template, and manage their profile.	through public registration (FR_30).
Expected frequency of use	Occasional, usually when a construction site affects their area or route.	Recurring, especially for subscribed or nearby construction sites.	Frequent: periodic data updates and daily review of reported comments.
Technical profile	General public with varied digital skills; accesses from mobile or desktop.	Same as the visitor.	Trained in the use of the administration functions and in reading the open data sources.
Related user stories	US-01, US-02, US-05, US-14, US-18, US-19, US-23, US-24	US-03, US-04, US-06, US-07, US-08, US-09, US-10, US-11, US-15, US-16, US-20, US-21, US-22	US-12, US-13, US-17, US-25

1.3 Business Rules

ID	Business rule	Related requirements
BR-01	Only registered and logged-in users can rate, comment, report comments, compare construction sites, export data, subscribe to updates and download the petition template.	FR_05, FR_06, FR_08, FR_09, FR_11, FR_14, FR_22
BR-02	A user must accept the personal data processing policy before their registration is completed; the acceptance and its date are recorded.	FR_17, NFR_13

BR-03	Each email address can be associated with only one user account.	FR_17
BR-04	Administrator accounts are created only by the system; public registration always creates a registered citizen account.	FR_30, NFR_15
BR-05	A registered user can give only one rating per construction site, with a value from 0 to 5 stars, and can update it. The construction site's satisfaction score is the average of its ratings.	FR_08
BR-06	A registered user can publish at most one comment per minute.	FR_09, NFR_04
BR-07	A user can delete only their own comments; the administrator can delete any comment.	FR_10, FR_12
BR-08	A user cannot report their own comments and can report the same comment only once.	FR_11
BR-09	A construction site is considered **active** when it has a start date, has not been marked as finished, and its physical progress is below 100%. Only active construction sites are shown on the map.	FR_01, FR_25
BR-10	The **performance index** of a construction site is the ratio between its actual physical progress and the progress planned for the current date. A value of 1 or more means the construction site is on schedule.	FR_04, FR_22, FR_23, FR_24
BR-11	Every construction site must record the source of its data and the date of its last update.	FR_26, NFR_14
BR-12	Each update of a construction site's physical progress is stored as a historical record; previous records are never overwritten, so that the evolution graph can be built.	FR_26, FR_27

BR-13	The system does not file rights of petition before public entities; it only provides a guide and a downloadable template that the user must complete and file directly with the competent entity.	FR_14, FR_15
BR-14	Subscribed users receive an email when the physical progress or the estimated end date of the construction site changes.	FR_06

2. CRC Cards

Users:

User *(abstract; specialized by Citizen and Administrator)*

Responsibilities	Collaborators
Knows its unique email (BR-03), hashed password (NFR_09), name, profile photo, description and role	NotificationService
Knows whether it accepted the personal data policy, and when (BR-02, NFR_13)	
Authenticates with its credentials (FR_16, FR_30)	
Updates its profile data and password (FR_20)	
Requests password recovery (FR_18)	

Citizen *(extends User)*

Responsibilities	Collaborators
Knows its ratings, comments, reports and subscriptions	ConstructionSite
Rates a construction site (FR_08)	Rating
Publishes a comment, at most one per minute (FR_09, BR-06)	Comment
Deletes its own comments (FR_10, BR-07)	CommentReport
Reports another user's comment (FR_11, BR-08)	Subscription
Subscribes to or unsubscribes from a construction site (FR_06)	PetitionTemplate
Shows its comment history (FR_21)	
Requests the right of petition template (FR_14)	

Administrator *(extends User)*

Responsibilities	Collaborators
Creates, edits and deletes construction sites (FR_26)	ConstructionSite
Registers progress updates of a construction site (BR-12)	ProgressRecord
Reviews reported comments	CommentReport
Deletes any comment (FR_12, BR-07)	Comment
Consults the dashboard of active construction sites (FR_25)	ConstructionSiteCatalog

Core domain:

ConstructionSite

Responsibilities	Collaborators
Knows its name, description, type, geometry (point or line), budget, start date, contractual end date, estimated end date, current physical progress and status (FR_01, FR_04)	Locality
Knows its data source and last update date (BR-11)	Contractor
Knows its locality, contractor and responsible entity	PublicEntity
Determines whether it is active (BR-09)	ProgressRecord
Calculates its performance index (BR-10)	RoadClosure
Calculates its average rating (BR-05)	SiteImage
Registers a new progress record without overwriting previous ones (BR-12)	Rating
Provides its progress history for the evolution graph (FR_27)	Comment
Notifies its subscribers when its progress or estimated end date changes (BR-14)	Subscription

ConstructionSiteCatalog *(non-persistent)*

Responsibilities	Collaborators
Lists the active construction sites to display on the map (FR_01, BR-09)	ConstructionSite
Searches construction sites by keyword in name or description (FR_19)	Locality
Filters by locality, physical progress, type and contractor (FR_07)	Contractor
Sorts by budget, physical progress, start date and estimated end date (FR_13)	
Paginates results, 20 per page (NFR_01)	
Summarizes active construction sites for the dashboard (FR_25)	

Locality

Responsibilities	Collaborators
Knows its name and its construction sites	ConstructionSite
Counts its active construction sites (FR_23)	Contractor
Calculates its average performance index (FR_23)	
Identifies the contractors with the most construction sites in it (FR_23)	

Contractor

Responsibilities	Collaborators
Knows its name, tax ID and its construction sites	ConstructionSite
Counts its construction sites (FR_24)	
Calculates its average performance index (FR_24)	

PublicEntity

Responsibilities	Collaborators
Knows its name, acronym, address and petition channel	ConstructionSite
Knows the construction sites it is responsible for	PetitionTemplate
Provides its recipient data for the petition template (FR_15)	

Construction site details:

ProgressRecord

Responsibilities	Collaborators
Knows its date, physical progress percentage and estimated end date at that moment	ConstructionSite
Knows which administrator registered it	Administrator
Validates that the progress is between 0 and 100	
Provides a data point for the evolution graph (FR_27)	

Milestone

Responsibilities	Collaborators
Knows its name, planned date and actual completion date	ConstructionSite
Determines whether it is completed (FR_28)	

RoadClosure

Responsibilities	Collaborators
Knows the affected road section (geometry), description, start date and end date	ConstructionSite
Determines whether it is in force on a given date (FR_29)	

SiteImage

Responsibilities	Collaborators

Knows its file, caption and date	ConstructionSite
Provides the image for the construction site gallery (FR_04)	

Citizen participation:

Rating

Responsibilities	Collaborators
Knows its score, date, author and construction site	Citizen
Validates that the score is between 0 and 5 (FR_08)	ConstructionSite
Ensures one rating per citizen per construction site, and allows updating it (BR-05)	

Comment

Responsibilities	Collaborators
Knows its text, publication date, author and construction site (FR_09)	Citizen
Verifies whether a given user can delete it (BR-07)	ConstructionSite
Knows the reports it has received	CommentReport

CommentReport

Responsibilities	Collaborators
Knows its reason, date, reporter, reported comment and status (pending or resolved)	Citizen

Validates that the reporter is not the author and has not reported that comment before (BR-08)	Comment
Marks itself as resolved after review	Administrator

Subscription

Responsibilities	Collaborators
Knows the subscribed citizen, the construction site and its creation date	Citizen
Sends an update to the citizen when the construction site changes (FR_06, BR-14)	ConstructionSite
	NotificationService

PetitionTemplate *(non-persistent)*

Responsibilities	Collaborators
Knows the required sections: recipient entity, requester contact, request, reasons, evidence and response channel (FR_15)	ConstructionSite
Pre-fills itself with the data of the construction site and its entity (FR_15)	PublicEntity
Includes the notice that it does not constitute an official filing (BR-13)	Citizen

Exports itself as a PDF (FR_14)	PdfGenerator

Services:

NotificationService *(non-persistent; uses the external email service)*

Responsibilities	Collaborators
Sends the password recovery email (FR_18)	User
Sends construction site update emails to subscribers (FR_06)	Subscription

PdfGenerator *(non-persistent)*

Responsibilities	Collaborators
Generates the PDF data sheet of a construction site (FR_05)	ConstructionSite
Generates the PDF of the right of petition template (FR_14)	PetitionTemplate

3. Class Diagram

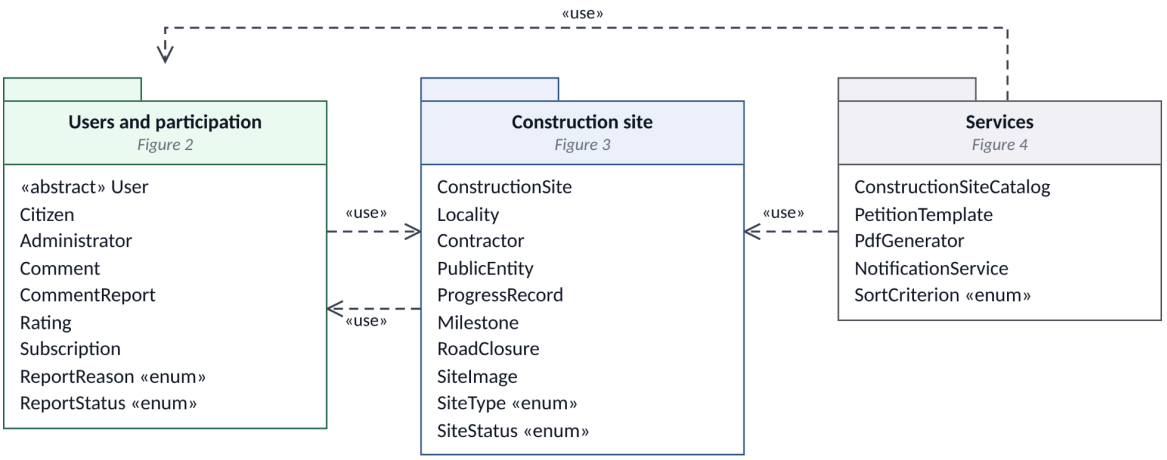


Figure 1

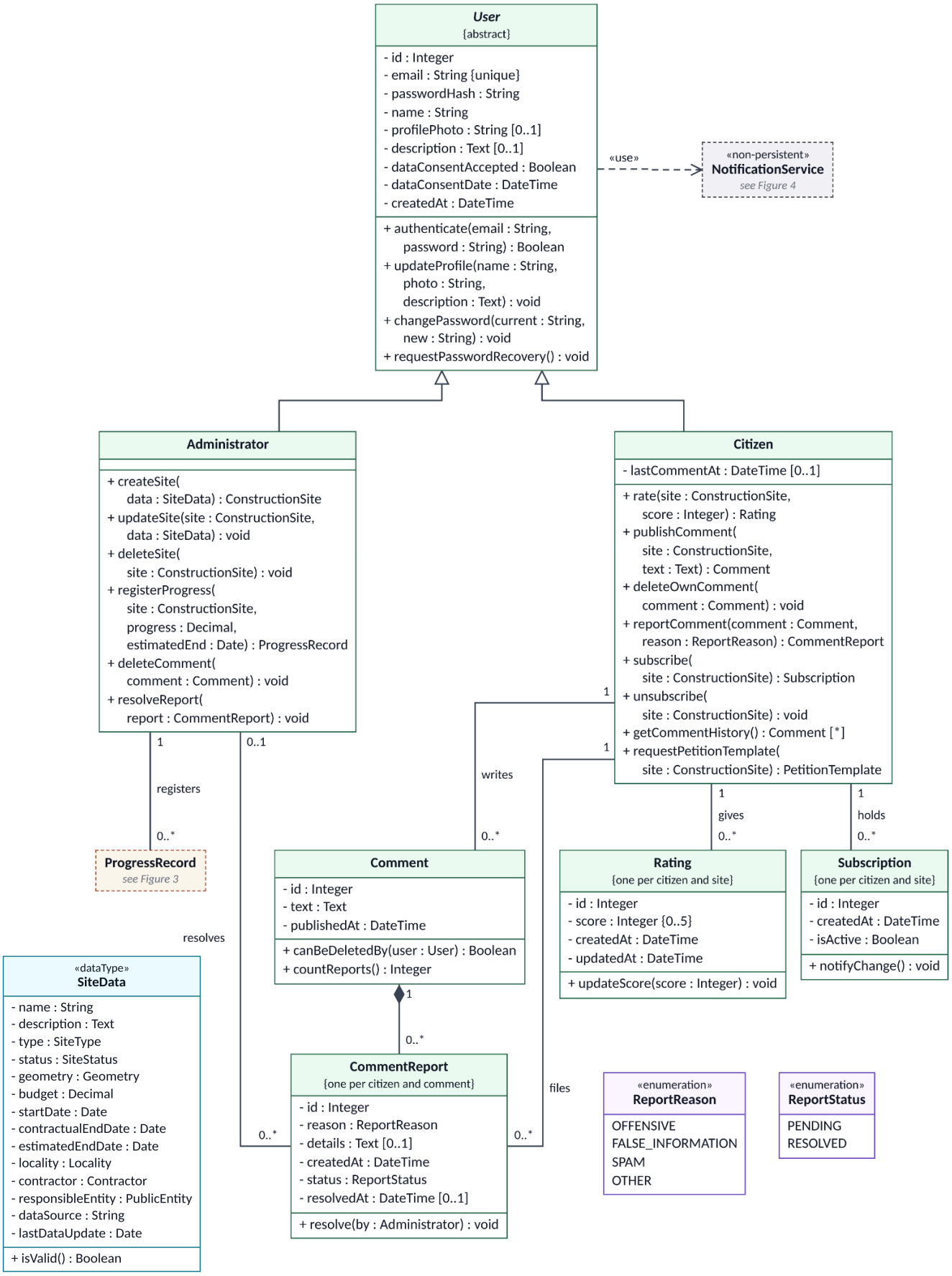

User
{abstract}
- id : Integer
- email : String {unique}
- passwordHash : String
- name : String
- profilePhoto : String [0..1]
- description : Text [0..1]
- dataConsentAccepted : Boolean
- dataConsentDate : DateTime
- createdAt : DateTime
+ authenticate(email : String, password : String) : Boolean
+ updateProfile(name : String, photo : String, description : Text) : void
+ changePassword(current : String, new : String) : void
+ requestPasswordRecovery() : void
«use»
«non-persistent»
NotificationService
see Figure 4
Administrator
+ createSite(data : SiteData) : ConstructionSite
+ updateSite(site : ConstructionSite, data : SiteData) : void
+ deleteSite(site : ConstructionSite) : void
+ registerProgress(site : ConstructionSite, progress : Decimal, estimatedEnd : Date) : ProgressRecord
+ deleteComment(comment : Comment) : void
+ resolveReport(report : CommentReport) : void
Citizen
- lastCommentAt : DateTime [0..1]
+ rate(site : ConstructionSite, score : Integer) : Rating
+ publishComment(site : ConstructionSite, text : Text) : Comment
+ deleteOwnComment(comment : Comment) : void
+ reportComment(comment : Comment, reason : ReportReason) : CommentReport
+ subscribe(site : ConstructionSite) : Subscription
+ unsubscribe(site : ConstructionSite) : void
+ getCommentHistory() : Comment [*]
+ requestPetitionTemplate(site : ConstructionSite) : PetitionTemplate
1
registers
0..*
0..1
writes
1
0..*
1
gives
0..*
1
holds
0..*
ProgressRecord
see Figure 3
Comment
- id : Integer
- text : Text
- publishedAt : DateTime
+ canBeDeletedBy(user : User) : Boolean
+ countReports() : Integer
Rating
{one per citizen and site}
- id : Integer
- score : Integer {0..5}
- createdAt : DateTime
- updatedAt : DateTime
+ updateScore(score : Integer) : void
Subscription
{one per citizen and site}
- id : Integer
- createdAt : DateTime
- isActive : Boolean
+ notifyChange() : void
resolves
1
0..*
files
0..*
0..*
«dataType»
SiteData
- name : String
- description : Text
- type : SiteType
- status : SiteStatus
- geometry : Geometry
- budget : Decimal
- startDate : Date
- contractualEndDate : Date
- estimatedEndDate : Date
- locality : Locality
- contractor : Contractor
- responsibleEntity : PublicEntity
- dataSource : String
- lastDataUpdate : Date
+ isValid() : Boolean
CommentReport
{one per citizen and comment}
- id : Integer
- reason : ReportReason
- details : Text [0..1]
- createdAt : DateTime
- status : ReportStatus
- resolvedAt : DateTime [0..1]
+ resolve(by : Administrator) : void
«enumeration»
ReportReason
OFFENSIVE
FALSE_INFORMATION
SPAM
OTHER
«enumeration»
ReportStatus
PENDING
RESOLVED

Figure 2

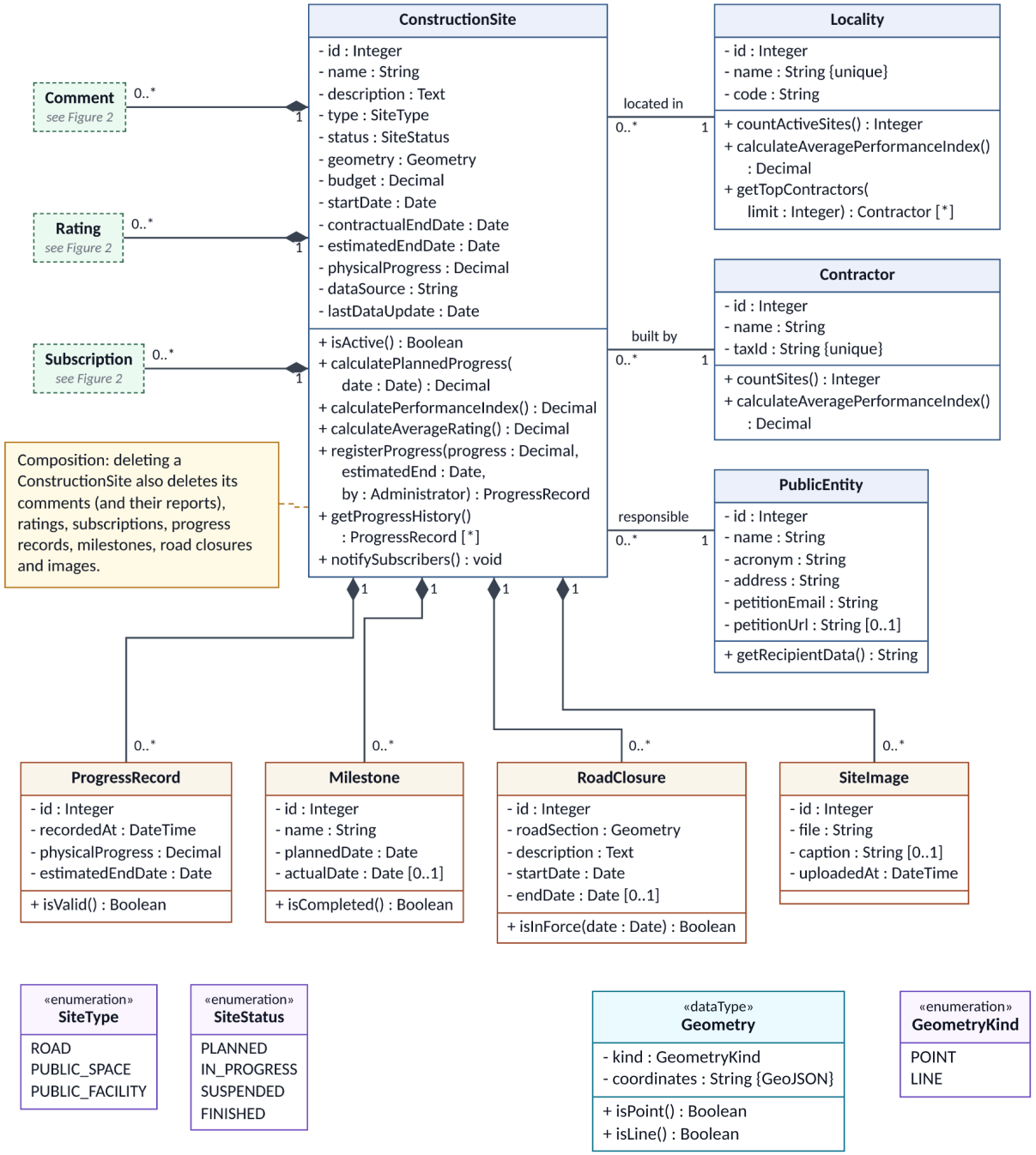

ConstructionSite
- id : Integer
- name : String
- description : Text
- type : SiteType
- status : SiteStatus
- geometry : Geometry
- budget : Decimal
- startDate : Date
- contractualEndDate : Date
- estimatedEndDate : Date
- physicalProgress : Decimal
- dataSource : String
- lastDataUpdate : Date
+ isActive() : Boolean
+ calculatePlannedProgress(date : Date) : Decimal
+ calculatePerformanceIndex() : Decimal
+ calculateAverageRating() : Decimal
+ registerProgress(progress : Decimal, estimatedEnd : Date, by : Administrator) : ProgressRecord
+ getProgressHistory() : ProgressRecord [*]
+ notifySubscribers() : void
Comment
see Figure 2
Rating
see Figure 2
Subscription
see Figure 2
Composition: deleting a ConstructionSite also deletes its comments (and their reports), ratings, subscriptions, progress records, milestones, road closures and images.
Locality
- id : Integer
- name : String {unique}
- code : String
+ countActiveSites() : Integer
+ calculateAveragePerformanceIndex() : Decimal
+ getTopContractors(limit : Integer) : Contractor [*]
located in
Contractor
- id : Integer
- name : String
- taxId : String {unique}
+ countSites() : Integer
+ calculateAveragePerformanceIndex() : Decimal
built by
PublicEntity
- id : Integer
- name : String
- acronym : String
- address : String
- petitionEmail : String
- petitionUrl : String [0..1]
+ getRecipientData() : String
responsible
ProgressRecord
- id : Integer
- recordedAt : DateTime
- physicalProgress : Decimal
- estimatedEndDate : Date
+ isValid() : Boolean
Milestone
- id : Integer
- name : String
- plannedDate : Date
- actualDate : Date [0..1]
+ isCompleted() : Boolean
RoadClosure
- id : Integer
- roadSection : Geometry
- description : Text
- startDate : Date
- endDate : Date [0..1]
+ isInForce(date : Date) : Boolean
SiteImage
- id : Integer
- file : String
- caption : String [0..1]
- uploadedAt : DateTime
«enumeration» SiteType
ROAD
PUBLIC_SPACE
PUBLIC_FACILITY
«enumeration» SiteStatus
PLANNED
IN_PROGRESS
SUSPENDED
FINISHED
«dataType» Geometry
- kind : GeometryKind
- coordinates : String {GeoJSON}
+ isPoint() : Boolean
+ isLine() : Boolean
«enumeration» GeometryKind
POINT
LINE

Figure 3

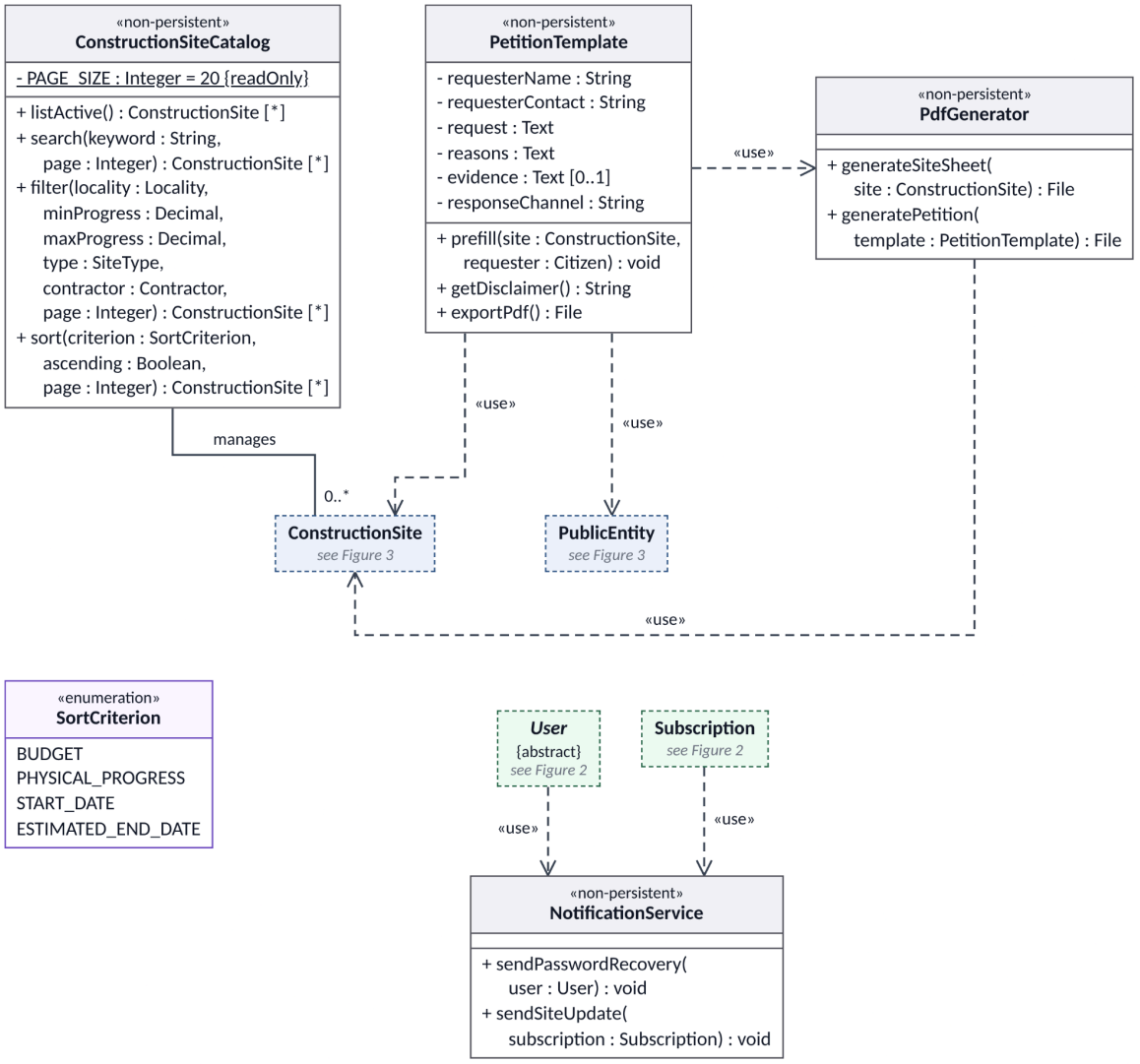

«non-persistent»
ConstructionSiteCatalog
- PAGE_SIZE : Integer = 20 {readOnly}
+ listActive() : ConstructionSite [*]
+ search(keyword : String,
page : Integer) : ConstructionSite [*]
+ filter(locality : Locality,
minProgress : Decimal,
maxProgress : Decimal,
type : SiteType,
contractor : Contractor,
page : Integer) : ConstructionSite [*]
+ sort(criterion : SortCriterion,
ascending : Boolean,
page : Integer) : ConstructionSite [*]
«non-persistent»
PetitionTemplate
- requesterName : String
- requesterContact : String
- request : Text
- reasons : Text
- evidence : Text [0..1]
- responseChannel : String
+ prefill(site : ConstructionSite,
requester : Citizen) : void
+ getDisclaimer() : String
+ exportPdf() : File
«non-persistent»
PdfGenerator
+ generateSiteSheet(
site : ConstructionSite) : File
+ generatePetition(
template : PetitionTemplate) : File
«use»
«use»
«use»
manages
0..*
«use»
ConstructionSite
see Figure 3
PublicEntity
see Figure 3
«enumeration»
SortCriterion
BUDGET
PHYSICAL_PROGRESS
START_DATE
ESTIMATED_END_DATE
User
{abstract}
see Figure 2
Subscription
see Figure 2
«use»
«use»
«non-persistent»
NotificationService
+ sendPasswordRecovery(
user : User) : void
+ sendSiteUpdate(
subscription : Subscription) : void

Figure 4

4. Relational Database Model

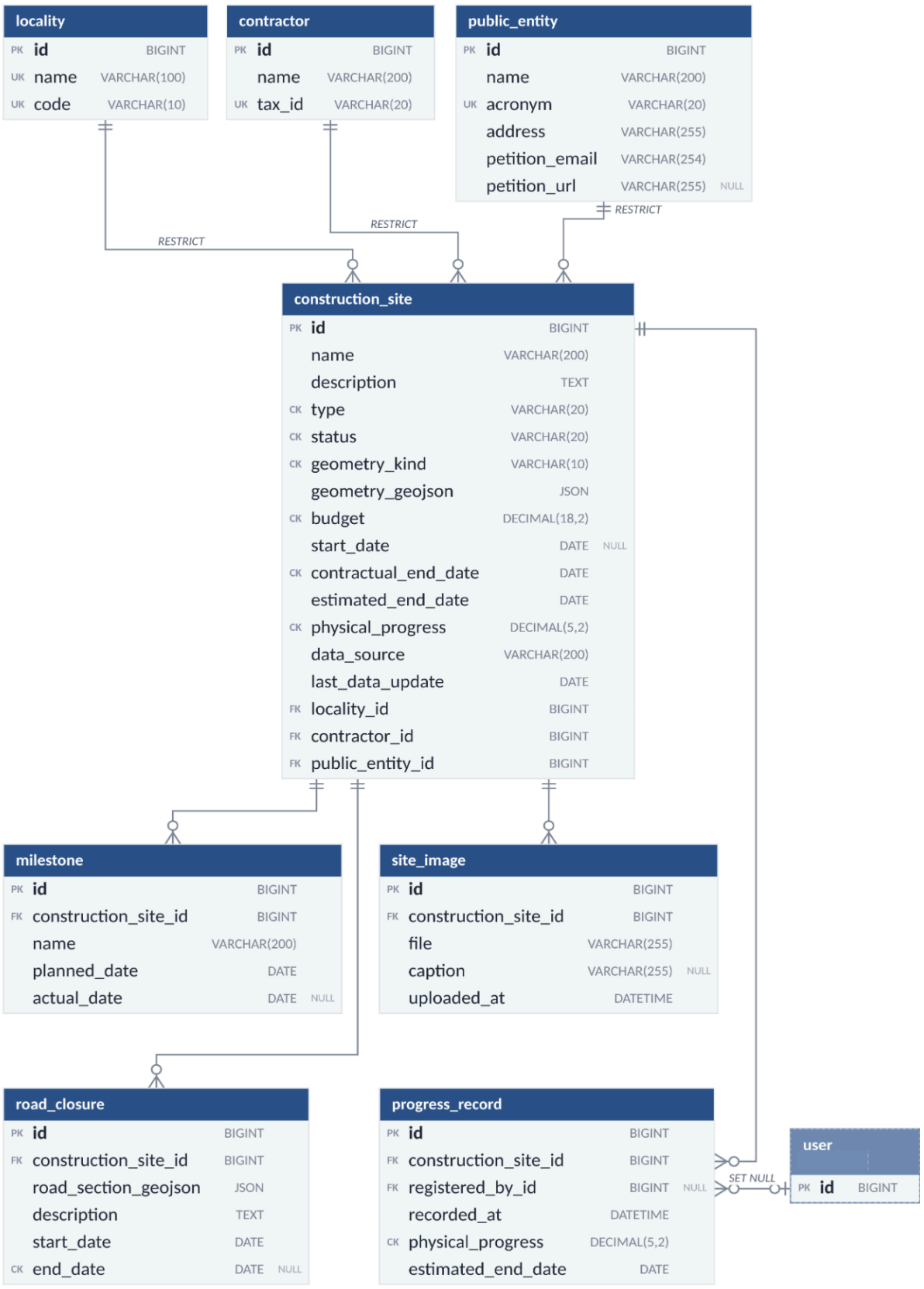


Figure 5


user
PK id BIGINT
UK email VARCHAR(254)
password_hash VARCHAR(128)
name VARCHAR(150)
profile_photo VARCHAR(255) NULL
description TEXT NULL
CK role VARCHAR(10)
CK data_consent_accepted BOOLEAN
data_consent_date DATETIME NULL
last_comment_at DATETIME NULL
created_at DATETIME
construction_site
PK id BIGINT
rating
PK id BIGINT
FK U1 citizen_id BIGINT
FK U1 construction_site_id BIGINT
CK score SMALLINT
created_at DATETIME
updated_at DATETIME
subscription
PK id BIGINT
FK U1 citizen_id BIGINT
FK U1 construction_site_id BIGINT
created_at DATETIME
is_active BOOLEAN
comment
PK id BIGINT
FK author_id BIGINT
FK construction_site_id BIGINT
text TEXT
published_at DATETIME
comment_report
PK id BIGINT
FK U1 comment_id BIGINT
FK U1 reporter_id BIGINT
FK resolved_by_id BIGINT NULL
CK reason VARCHAR(20)
details TEXT NULL
CK status VARCHAR(10)
created_at DATETIME
resolved_at DATETIME NULL
SET NULL

Figure 6

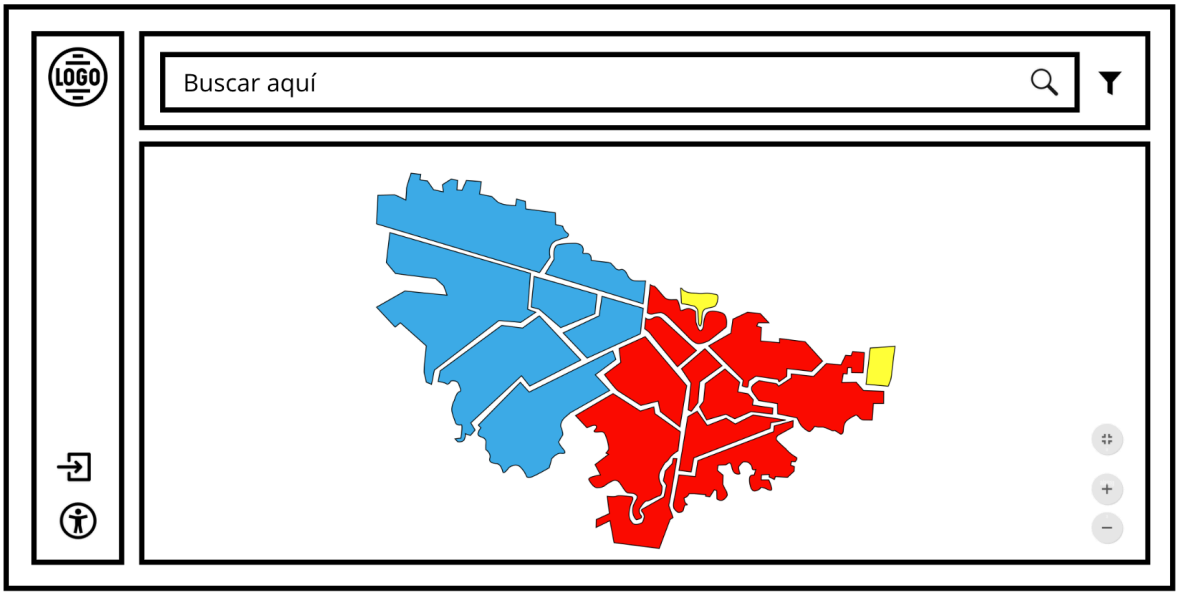
CHECK Restrictions

Tabla.columna	Restricción
construction_site.type	IN (‘ROAD’, ‘PUBLIC_SPACE’, ‘PUBLIC_FACILITY’)
construction_site.status	IN (‘PLANNED’, ‘IN_PROGRESS’, ‘SUSPENDED’, ‘FINISHED’)
construction_site.geometry_kind	IN (‘POINT’, ‘LINE’)
comment_report.reason	IN (‘OFFENSIVE’, ‘FALSE_INFORMATION’, ‘SPAM’, ‘OTHER’)
comment_report.status	IN (‘PENDING’, ‘RESOLVED’)
user.role	IN (‘CITIZEN’, ‘ADMIN’)
user.data_consent_accepted	role = ‘ADMIN’ OR data_consent_accepted = TRUE (BR-02)
construction_site.physical_progress, progress_record.physical_progress	BETWEEN 0 AND 100
construction_site.budget	≥ 0
construction_site.contractual_end_date	≥ start_date
road_closure.end_date	IS NULL OR ≥ start_date
rating.score	BETWEEN 0 AND 5 (BR-05)

5. Mockups

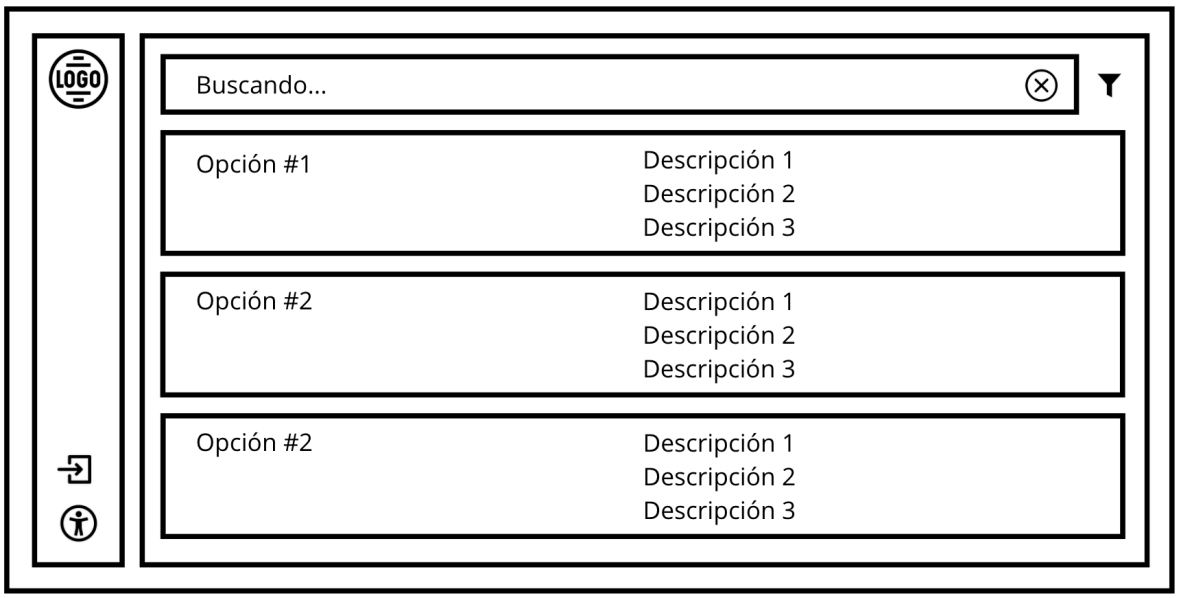
1. Main Page

From the main page, users can navigate the map or use the search engine to locate active public works, log in with their credentials, or customize accessibility settings.



2. Search

Users can search for active public works by name or address. Additionally, the filter option allows users to refine their search criteria for more specific results.



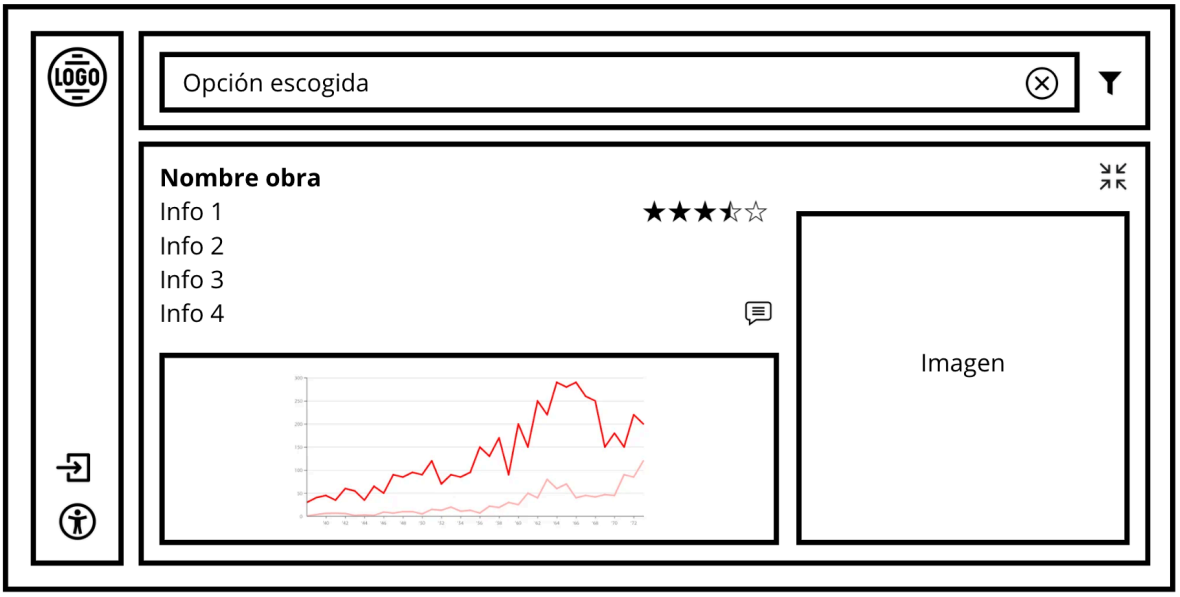
3. Map

When the user selects one of the active public works, the system will show a popup window with basic information of the selected work.



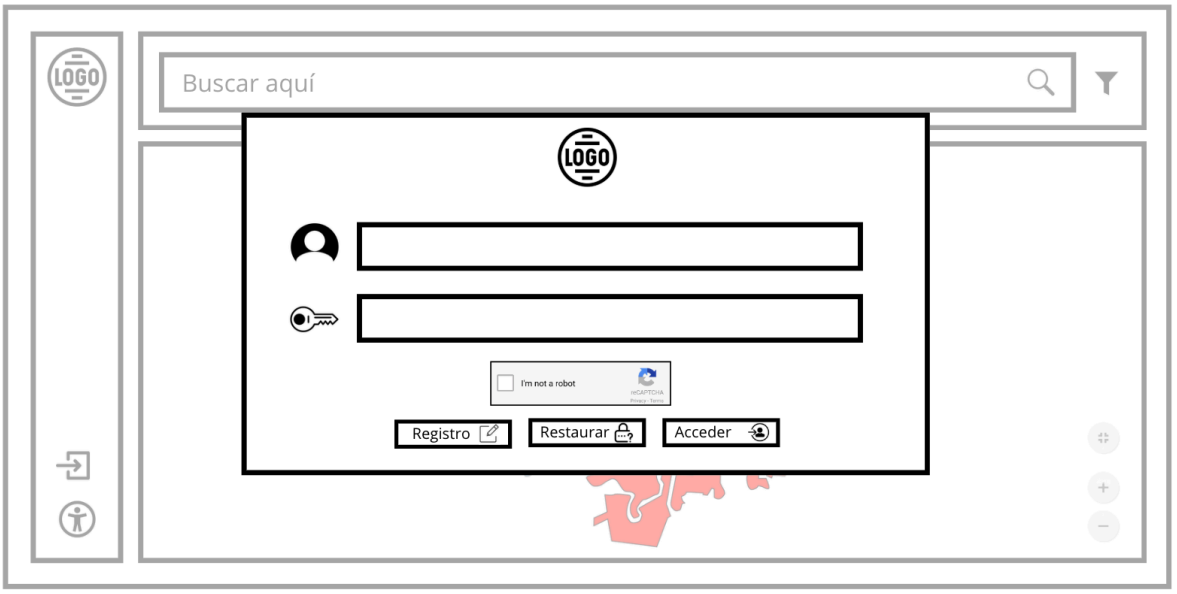
4. Expand information window

When the user clicks the expand window icon, then the system will open the Work information module showing information, rates, comments, progress graph and image of the selected work.



5. Login window

Given that the user desires to access his profile, when the user selects the login icon, then the login popup window will appear, so the user can enter his credentials and select the login button.



4. Profile page

When the user presses the profile icon, then the system will open the profile window, where the user can access his profile information, saved works on the favorite tab, lawsuits and credentials.

